Sta301

Final Term (Live Quiz)

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Question 1:
In interval estimation we obtained a ……of values as an estimate of parameter.

- ✕ None of these
- ✓ Range
- ✕ Both a and b
- ✕ Group

Question 2:
A statistic whose standard error decreases with an increase in the sample size will be ……

- ✕ Consistant
- ✓ Efficient
- ✕ None
- ✕ Unbiased

Question 3:
As a rule of thumb, when n>=30, then we can assume that……..is normally distributed:

- ✕ Probability distribution
- ✕ Binomial distribution
- ✕ Sampling distribution
- ✓ Sampling distribution of sample mean

Question 4:
In repeating sampling, which of the following possess desirable property of being Unbiased?

- ✕ Sample median
- ✓ All of above
- ✕ Sample Mean
- ✕ Sample proportion

Question 5:
..... is used when we are dealing with proportions to find the probabilities:

- ✓ Normal distribution
- ✕ Both Z & t distribution
- ✕ F distribution
- ✕ t- distribution

Question 6:
The standard error of the sampling distribution of the sample mean when sampling is done with replacement is equal to:

- ✕ б
- ✕ s / (n - 1)
- ✓ б/√n
- ✕ s / n

Question 7:
The sampling distribution of the sample proportion, p^, will be _____ distributed if the sample size is large and p^ is not too close to 0 or l:

- ✕ Normal
- ✓ Approximately normal
- ✕ Poisson
- ✕ Binomial

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Question 8:
In case of sampling with replacement from two infinite populations, the standard deviation of the sampling distribution of bar X1 - bar X2 standard error of bar X1 - bar X2 denoted by:

- ✗ 1/n
- ✗ √((σ1+σ2)/n)
- ✓ √((σ1)²/n1) + (σ2)²/n2
- ✗ √((σ1-σ2)/n)

Question 9:
Providing as an estimate a single value calculated from the sample that is likely to be close in magnitude to the unknown parameter is known as:

- ✗ None of the above
- ✗ Estimation
- ✗ Interval Estimation
- ✓ Point Estimation

Question 10:
Which of the following relationships exist between sampling distribution of sample mean and sampling distribution of sample median?

- ✓ Var(mean) < Var(median)
- ✗ None of these
- ✗ Var(median) < Var(median)
- ✗ Both of these

Question 1: Which of the following is desirable for a good point estimator?

- Consistency
- Efficiency
- Unbiasedness
- **All of these ✓**

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Question 2: The end points that bound the confidence interval are called;

- Upper limits
- Bounded limits
- **Lower and upper limits ✓**
- Lower limits

Question 3: In case of sampling with replacement from two infinite populations, the standard deviation of the sampling distribution of bar X1 - bar X2 standard error of bar X1 - bar X2 denoted by:

- √((б1-б2)/n)
- √((б1+б2)/n)
- **√((б1)²/n1) + (б2)²/n2 ✓**
- 1/n

Question 4: In the method of moments, how many equations are required for finding two unknown population parameters?

- **2 ✓**
- 4
- 1
- 3

Question 5: In sampling from a large population with б= 25 . The size of the sample used is 10, the value of the standard error will be:

- 2
- **7.9 ✓**
- 0
- 6

Question 6: Which of the following statements about confidence intervals is inaccurate?

- If the population standard deviation increases, the confidence interval decreases in width

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- If we keep the confidence coefficient fixed, the confidence interval gets narrower as we increase the sample size
- A confidence interval for a mean always contains the sample mean
- **If we keep the sample size fixed, the confidence interval gets wider as we increase the confidence coefficient ✓**

Question 7: Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

- 325
- 216 ✓
- 256
- 196

Question 8: The central limit theorem states that the mean of the sampling distribution of the means is equal to the population:

- **Mean ✓**
- Random Error
- Standard error
- Standard deviation

Question 9: In repeating sampling, which of the following possess desirable property of being Unbiased?

- Sample proportion
- All of above ✓
- Sample Mean
- Sample median

Question 10: Sampling distributions are:

- Refers to the margin of error
- The probability distribution that is assigned to population parameters
- **The probability distribution of sample statistic ✓**

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- None of above

Question 1: The standard error of the sampling distribution of differences of means ($x_1 - x_2$), is equal to:

- None of the above
- $\sqrt{((б_1)^2/n_1) + (б_2)^2/n_2}$ ✅
- $\sqrt{(б/n)}$
- $\sqrt{((б_1+б_2)/n)}$

Question 2: "To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as:

- The method of least square
- The method of moments
- The method of fractional moments
- The method of maximum likelihood ✅

Question 3: In sampling from a large population with б = 60. The size of the sample used is 100, the value of the standard error will be:

- 4
- 15
- 6 ✅
- 42

Question 4: The mean of the sampling distribution of $\bar{x}_1 - \bar{x}_2$, denoted by $\mu\bar{x}_1 - \bar{x}_2$ is equal to the difference between respective:

- Population Proportions
- Population Means ✅
- Sample Means
- Sample Proportions

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Question 5: A statistic whose standard deviation decreases with an increase in sample size will be called:

- Sufficient
- Consistent
- Unbiased
- Efficient ✅

Question 6: In sampling from a large population with $\sigma = 35$, the standard error of the mean is found to be 4. The size of the sample used is:

- 77
- 56
- 25 ✅
- 15

Question 7: Ideally, the width of confidence interval should be:

- 100
- 0 ✅
- 98
- 1

Question 8: In general sample variance is a/an ……..estimator?

- Consistent
- Biased
- none of these
- Unbiased ✅

Question 9: In general, the estimators obtained by the method of moments are ……:

- Inefficient ✅
- Inconsistent
- Biased

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- All the above

Question 10: Confidence interval is also known as:

- Point estimator
- None of these
- Level of Significance
- Margin of error ☑

Question 1:
An estimator is said to be efficient if it has:

- None of these
- Both (a) & (b)
- Smallest Variance ✓
- Unbiased estimate

Question 2:
Sample variance is a/an_____estimator of population variance.

- None of these
- Biased
- Unbiased ✓
- Biased and Unbiased

Question 3:
The sampling distribution of the sample proportion, p̂, will be _____ distributed if the sample size is large and p̂ is not too close to 0 or 1:

- Poisson
- Binomial
- Approximately normal ✓
- Normal

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Question 4:
In general, the estimators obtained by the method of moments are …….

- Inconsistent
- Biased
- All the above
- Inefficient ✓

Question 5:
The standard error of the sampling distribution of $p_1 - p_2$ is equal to:

- $\sqrt{(p/n)}$
- $\sqrt{(p_1q_1/n_1) + p_2q_2/n_2}$ ✓
- None of the above
- $\sqrt{((p_1+p_2)/n)}$

Question 6:
In sampling without replacement; fpc is not used when:

- None of the above
- $n > 0.05N$
- $n \leq 0.05N$ ✓
- $n = 0.05N$

Question 7:
If our sampled population is normal, then sampling distribution of the sample mean will ……:

- None of the above
- Depends on sample size
- Normal ✓
- Not necessarily normal

Question 8:
If an estimator is more efficient than the other estimator, its shape of the sampling distribution will be:

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- Highly peaked ✓
- Normal
- Flattered
- Skewed to right

Question 9:
"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

- Critical value
- Critical region
- Confidence interval ✓
- Acceptance region

Question 10:
A confidence interval has a specified probability_____of containing the true value of the parameter.

- 1 - α ✓
- 1 + α
- α/2
- α

Question 1:
What does the central limit theorem state?

- The sampling distribution of the sample mean will be skewed
- The sample mean will always equal the population mean
- The sampling distribution of the sample mean approaches a normal distribution as sample size increases
- Sample size has no effect on the sampling distribution of the sample mean

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Question 2:
Which of the following is a measure of central tendency?

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- Variance
- Mean
- Standard deviation
- Range

Question 3:
In a standard normal distribution, what is the value of the mean?

- 0
- 1
- -1
- 2

Question 4:
Which of the following is true about a population parameter?

- It is calculated from a sample
- It is a summary measure of a population
- It varies from sample to sample
- It is always known

Question 5:
The probability that an event occurs is a number between:

- -1 and 1
- 0 and 1
- 0 and infinity
- -infinity and infinity

Question 6:
What does a p-value represent in hypothesis testing?

- The probability that the null hypothesis is true
- The probability that the sample data came from the population

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- The probability of obtaining a sample statistic at least as extreme as the one observed, assuming the null hypothesis is true
- The probability of rejecting the null hypothesis

Question 7:
Which of the following is an example of a continuous random variable?

- Number of students in a class
- Temperature in a city
- Number of cars in a parking lot
- Number of heads in a coin toss

Question 8:
Which of the following is used to test hypotheses about population means when the population standard deviation is unknown?

- Z-test
- T-test
- Chi-square test
- F-test

Question 9:
Which of the following is NOT a characteristic of a binomial distribution?

- There are two possible outcomes in each trial
- The trials are dependent
- The probability of success is constant across trials
- The number of trials is fixed

Question 10:
If the population distribution is normal, the sampling distribution of the sample mean will be:

- Always normal
- Always skewed
- Normal for sample sizes greater than 30

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- Skewed for sample sizes greater than 30

Question 11:
The standard deviation of a sample is usually denoted by:

- σ
- μ
- s
- $\bar{x}$

Question 12:
Which of the following is a correct interpretation of a 95% confidence interval for the mean?

- There is a 95% chance the population mean lies within the interval
- We are 95% confident the sample mean lies within the interval
- 95% of all possible sample means fall within the interval
- If we take many samples, 95% of the resulting confidence intervals will contain the population mean

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Question 13:
Which of the following is true about the correlation coefficient (r)?

- It ranges from -1 to 1
- It can be any real number
- It indicates the slope of a regression line
- A correlation of 0 indicates a perfect positive relationship

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Question 14:
A random variable that can take any value in a given range is called a:

- Discrete random variable
- Continuous random variable
- Nominal random variable
- Ordinal random variable

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Question 15:
What does a high variance in a dataset indicate?

- The data points are very close to the mean
- The data points are very spread out around the mean
- The mean is close to the median
- The dataset has no variation

Question 16:
Which of the following distributions is used to test the goodness of fit?

- Normal distribution
- Chi-square distribution
- T-distribution
- Exponential distribution

Question 17:
In a hypothesis test, the null hypothesis (H0) is:

- The hypothesis that is always accepted
- The hypothesis that is tested and possibly rejected
- The alternative hypothesis
- Always true

Question 18:
Which of the following is a key assumption of the t-test?

- The sample size must be greater than 30
- The population distribution must be normal or nearly normal
- The population variance must be known
- The sample must be selected non-randomly

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Question 19:
What is the term for the difference between the observed value and the expected value in a chi-square test?

- Error term
- Residual
- Deviation
- Test statistic

Question 20:
If a test statistic falls in the critical region, what should be the decision?

- Fail to reject the null hypothesis
- Reject the null hypothesis
- Accept the alternative hypothesis
- The test is inconclusive

The confidence intervals become wide and less precise by:

- None of these
- decreasing the level of significance
- increasing the sample size
- ✓ decreasing the sample size

A random sample of size n = 25 is drawn from a normal population with population mean = 50 and sigma = 10. Then mean of the sampling distribution of the sample mean is:

- ✓ 50
- 2
- 25
- 45

In repeating sampling, which of the following possess desirable property of being Unbiased?

- Sample median
- Sample proportion
- All of above

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- ✓ Sample Mean

In interval estimation we obtained a ……of values as an estimate of parameter.

- None of these
- Both a and b
- Group
- ✓ Range

An estimator is said to be……. if its expected value is equal to true value of its parameter.

- Consistant
- ✓ Unbiased
- Efficient
- Biased

The probability distribution of the proportion of successes in all possible samples is called the:

- Sampling distribution of Population proportions
- Sampling probability distribution
- ✓ Sampling distribution of sample proportions
- Sampling distribution

Sample Proportion is a /an_____ estimator of population mean.

- ✓ Unbiased and Consistent estimator
- Consistent
- None of these
- Unbiased

If our sampled population is normal, then sampling distribution of the sample mean will ……:

- Not necessarily normal
- ✓ Normal
- None of the above
- Depends on sample size

A parameter is a ….. quantity.

- Random

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- ✓ Constant
- Sample
- Variable

 The end points that bound the confidence interval are called:

- ✓ Lower and upper limits
- Lower limits
- Bounded limits
- Upper limits

Question # 1
If Var(T1) > Var(T2), where T1 and T2 are two unbiased estimators, then:

- T2 is more efficient
- ☑ T1 is more efficient
- T2 is more consistent
- T1 is more consistent

Question # 2
In sampling from a large population with σ = 45. The size of the sample used is 12, the value of the standard error will be:

- 12.99
- 1.2
- ☑ 3.66
- 5.9

Question # 3
Which of the following statements about confidence intervals is inaccurate?

- A confidence interval for a mean always contains the sample mean
- If the population standard deviation increases, the confidence interval decreases in width
- If we keep the confidence coefficient fixed, the confidence interval gets narrower as we increase the sample size

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- ✅ If we keep the sample size fixed, the confidence interval gets wider as we increase the confidence coefficient

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Question # 4
A consistent estimator will always be a:

- Biased estimator
- Unbiased estimator and biased estimator
- ✅ May or may not be unbiased estimator
- Unbiased estimator

Question # 5
Simple random sampling is appropriate when the units in the population are:

- Not accessible
- ✅ Homogeneous
- Accessible
- Heterogeneous

Question # 6
“A point estimate plus/minus a few times the standard error of that estimate”. This statement represents:

- Acceptance region
- ✅ Confidence interval
- Critical value
- Critical region

Question # 7
P: 0 1/3 2/3 1
f(p): 1/10 4/10 4/10 1/10

The mean of the sampling proportion will be:

- 1/15
- 10/20

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- 1/20
- ✅ 15/30

Question # 8
What is the rule of thumb for a sample size "n" to be considered large?

- ✅ n=30
- None of the above
- n>30 or n=30
- n<30

Question # 9
In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- 40
- 100
- 10
- ✅ 20

Question # 10
In a Poisson distribution, Maximum Likelihood Estimator (MLE) for μ (mu) is equal to:

- Reciprocal of the sample variance
- Sample variance
- Reciprocal of the mean
- ✅ Sample mean

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Question 1:
An industrial designer wants to determine the average amount of time it takes an adult to assemble an “easy to assemble” toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is a 95% confidence interval for the mean assembly time?

- (17.1123, 24.7277)

- (17.1123, 30.7277)
- (14.1123, 22.7277)
- **(17.1123, 22.7277)** ✅

Question 2:
If Var(T1) > Var(T2), where T1 and T2 are two unbiased estimators, then:

- T1 is more consistent
- T1 is more efficient
- T2 is more consistent
- **T2 is more efficient** ✅

Question 3:
The sampling distribution of the sample proportion, p^, will be _____ distributed if the sample size is large and p^ is not too close to 0 or 1:

- Poisson
- **Approximately normal** ✅
- Normal
- Binomial

Question 4:
What condition is required before the central limit theorem justifies approximating the sampling distribution of the mean with a normal distribution?

- 20n < N
- **n > 30** ✅
- n(1 - p) > 5
- n < 30

Question 5:
If sampling is from a normal population, the mean of the sampling distribution of the sampled means is equal to:

- б/√n

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- s/(n-1)
- **The mean of the sampled population** ✅
- sigma the standard deviation of the population

Question 6:
When an interval is constructed, what is the probability of occurrence of parameter in it?

- **Either 1 or 0** ✅
- 1
- 0
- 50

Question 7:
The precision of an estimator can be increased by decreasing the:

- **Size of sample** ✅
- Level of significance
- Number of parameters
- Number of estimators

Question 8:
Sample mean is a/an ..... estimate of population mean.

- **Unbiased** ✅
- Biased
- Either unbiased or biased
- None of the above

Question 9:
The difference between the value of the statistic and the value of the parameter is called _____.

- Random Error
- Standard Error

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- Biased Error
- **Sampling Error** ☑

Question 10:
In sampling from a large population with б = 60. The size of the sample used is 100, the value of the standard error will be:

- **6** ☑
- 42
- 4
- 15

Question 1:
Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

- 10 ✔
- 18
- 15
- 20

Question 2:
The mean of the sampling proportion will be:

- 11/20
- 12/20
- 9/20 ✔
- 10/20

Question 3:
If sampling is from a normal population, the mean of the sampling distribution of the sampled means is equal to:

- б/√n
- sigma the standard deviation of the population
- The mean of the sampled population ✔

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- s/(n-1)

Question 4:
Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

- 3.568
- 2.500 ✔
- 1.654
- 3.125

Question 5:
In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- 100 ✔
- 10
- 20
- 40

Question 6:
The standard error of the sampling distribution of p1 - p2 is equal to:

- √((p1 + p2)/n)
- None of the above
- √(p/n)
- √(p1q1/n1) + p2q2/n2 ✔

Question 7:
The limitation of point estimation is that:

- All of these
- There is no way of ascertaining how close the point estimate to parameter? ✔
- It is not calculated for small sample
- It is not calculated for non-normal distribution

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Question 8:
If P (6.2 < ? < 12.2) = 0.95 then:

- We are 5% confident that true value of the parameter fall in this interval
- We are 95% confident that true value of the parameter fall in this interval ✔
- All are correct
- This is wrong statement

Question 9:
Which of the following is most important and most widely used method in point estimation?

- The method of fractional moments
- The method of least square
- The method of moments
- The method of maximum likelihood ✔

Question 10:
If our sampled population is normal, then sampling distribution of the sample mean will ……:

- Normal ✔
- None of the above
- Not necessarily normal
- Depends on sample size

Question #1:
Which of the following is desirable for a good point estimator?

- **Consistency**
- **Efficiency**
- **Unbiasedness**
- **All of these** ☑

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Question #2:
A random sample of n = 6 has the elements 6, 10, 13, 14, 18, and 20. What is the point estimate of the population mean?

- **11**
- **12**
- **13.5** ✓
- **11.5**

Question #3:
As sample size goes up, what tends to happen to 95% confidence intervals?

- **They become more precise and narrow** ✓
- **They become more precise**
- **They become more narrow**
- **They become wider**

Question #4:
Suppose we have a population of size N = 5 i.e. 0, 11, 20, 23, 45, and if we draw all possible samples of size n = 2 with replacement then total possible samples will be:

- **25** ✓
- **10**
- **50**
- **35**

Question #5:
The precision of an estimator can be increased by increasing the;

- **Size of sample** ✓
- **Size of population**
- **Number of parameters**
- **Number of estimators**

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Question #6:
A parameter is a ..... quantity.

- **Constant** ✓
- **Random**
- **Sample**
- **Variable**

Question #7:
The mean of the sampling distribution of difference of means is represented by:

- **μ1 - μ2** ✓
- **None of above**
- **μ**
- **б**

Question #8:
In sampling from a large population with sigma = 35, the standard error of the mean is found to be 4. The size of the sample used is:

- **25** ✓
- **77**
- **15**
- **56**

Question #9:
In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- **40**
- **10**
- **20**
- **100** ✓

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Question #10:
The standard error of the sampling distribution of p1 - p2 is equal to:

- **√(p/n)**
- **√(p1q1/n1) + p2q2/n2** ✓
- **None of the above**
- **√((p1 + p2)/n)**

Question # 1: In sampling from a large population with sigma = 35, the standard error of the mean is found to be 4. The size of the sample used is:

- A) 15
- B) 25 ✓
- C) 77
- D) 56

Question # 2: The confidence intervals become wide and less precise by:

- A) None of these
- B) Decreasing the level of significance
- C) Increasing the sample size
- D) Decreasing the sample size ✓

Question # 3: When sampling is from an infinite population, the mean and standard error of the sampling distribution of p^ are _____ and_____, respectively:

- A) p, sqrt [p(1 - p)/n] ✓
- B) p, p(1 - p)
- C) p, n(1 - p)p
- D) p^, p(1 - p) / n

Question # 4: P 0 1/3 2/3 1
f(p) 1/20 9/20 9/20 1/20
The mean of the sampling proportion will be:

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- A) 11/20
- B) 9/20 ✅
- C) 12/20
- D) 10/20

Question # 5: If a statistic used as an estimator has its expected value equal to the true value of the population parameter being estimated, then it is called:

- A) Unbiased ✅
- B) Efficient
- C) Consistent
- D) Sufficient

Question # 6: A randomly selected sample of 400 students at a university with 15-week semesters was asked whether or not they think the semester should be shortened to 14 weeks. 46% of the 400 students surveyed answered "yes". Which one of the following statements about the number 46% is correct?

- A) It is a population parameter
- B) It is a standard error
- C) It is a margin of error
- D) It is a sample statistic ✅

Question # 7: An estimator is said to be efficient if it has:

- A) None of these
- B) Smallest Variance ✅
- C) Both (a) & (b)
- D) Unbiased estimate

Question # 8: In case of sampling with replacement from two infinite populations, the standard deviation of the sampling distribution of bar X1 - bar X2 (standard error of bar X1 - bar X2) is denoted by:

- A) $\sqrt{((б1)^2/n1) + (б2)^2/n2}$ ✅

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- B) √((б1 + б2)/n)
- C) 1/n
- D) √((б1 - б2)/n)

Question # 9: A standard deviation obtained from the sampling distribution of sample statistics is known as:

- A) Minimum error
- B) Standard error ✅
- C) Sampling error
- D) Universal error

Question # 10: If Var(T1) > Var(T2), where T1 and T2 are two unbiased estimators, then:

- A) T1 is more efficient
- B) T1 is more consistent
- C) T2 is more efficient ✅
- D) T2 is more consistent

Question 1:
Which of the following is not a criterion for a good problem situation?

- It must be authentic.
- It must be appropriate to the students' level of development.
- ✓ It must be such that its solution will be benefited by group effort.
- It must be clearly structured and defined.

Question 2:
Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

- 15
- ✓ 10
- 20
- 18

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Question 3:
In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- 40
- 10
- 20
- ✓ 100

Question 4:
We can apply method of Maximum Likelihood on:

- Continuous variables only
- Discrete variables only
- ✓ Discrete as well as continuous variables
- Qualitative variables only

Question 5:
An estimator is said to be efficient if it has:

- Both (a) & (b)
- None of these
- Unbiased estimate
- ✓ Smallest Variance

Question 6:
As a rule of thumb, when n>=30, then we can assume that……is normally distributed:

- Binomial distribution
- ✓ Sampling distribution of sample mean
- Probability distribution
- Sampling distribution

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Question 7:
If Var(mean1) < Var(mean2) it implies:

- Mean1 is more consistent
- ✓ Mean1 is more efficient
- Mean2 is more consistent
- Mean2 is more efficient

Question 8:
A consistent estimator will always be a:

- Unbiased estimator
- Unbiased estimator and biased estimator
- Biased estimator
- ✓ May or may not be unbiased estimator

Question 9:
The larger the n, the smaller the standard error, and so the narrower the:

- Confidence level
- Level of significance
- ✓ Confidence Interval
- None of the above

Question 10:
Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

- 3.568
- ✓ 2.500
- 1.654
- 3.125

Question 1:
As a rule of thumb, when n ≥ 30, then we can assume that……is normally distributed:

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- Sampling distribution
- Probability distribution
- **Sampling distribution of sample mean ✓**
- Binomial distribution

Question 2:
In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- **100 ✓**
- 20
- 40
- 10

Question 3:
A random sample of n = 6 has the elements 6, 10, 13, 14, 18, and 20. What is the point estimate of the population mean?

- 11.5
- 11
- **12 ✓**
- 13.5

Question 4:
In sampling from a large population with sigma = 35, the standard error of the mean is found to be 4. The size of the sample used is:

- 56
- 77
- 15
- **25 ✓**

Question 5:
Sample mean is a/an ..... estimate of population mean.

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- **Unbiased ✓**
- Biased
- Either unbiased or biased
- None of the above

Question 6:
A statistic whose standard error decreases with an increase in the sample size will be .....

- Unbiased
- **Efficient ✓**
- Consistent
- None

Question 7:
Suppose we have a population of size N = 5 i.e. 0, 11, 20, 23, 45 and if we draw all possible samples of size n = 2 with replacement then total possible samples will be:

- 35
- **25 ✓**
- 10
- 50

Question 8:
Which of the following must not be a consistent estimator?

- Sample variance
- Sample mean
- Sample proportion
- **Sample median ✓**

Question 9:
In sampling without replacement; fpc is not used when:

- **$n \leq 0.05N$ ✓**

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- n = 0.05N
- n > 0.05N
- None of the above

Question 10:
The mean of the sampling distribution of x¯1−x¯2\bar{x}_1 - \bar{x}_2x¯1−x¯2, denoted by μx¯1−x¯2\mu_{\bar{x}_1 - \bar{x}_2}μx¯1−x¯2 is equal to the difference between respective:

- Sample proportions
- **Population Means ✓**
- Sample Means
- Population Proportions

Question 1: In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

- 100 ✓
- 20
- 40
- 10

Question 2: Simple random sampling is appropriate, when the units in population are:

- Homogeneous ✓
- Not accessible
- Heterogeneous
- Accessible

Question 3: How can we interpret the 90% confidence interval for the mean of the normal population?

- There are 100% chances of falling true value of the parameter
- There are 10% chances of falling true value of the parameter
- All are correct
- There are 90% chances of falling true value of the parameter ✓

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Question 4: The sampling distribution of the mean becomes approximately normally distributed only when the following condition is met:

- The sample size is large. ✓
- The population is normally distributed.
- A single random sample is drawn from the population.
- The standard deviation of the population is large.

Question 5: In general, the estimators obtained by the method of moments are:

- All the above
- Biased ✓
- Inconsistent
- Inefficient

Question 6: The total number of possible samples of size 3 (with replacement) from the population of size 6 will be:

- 256
- 216 ✓
- 196
- 325

Question 7: If sampling is from a normal population, the mean of the sampling distribution of the sample means is equal to:

- The mean of the sampled population ✓
- sigma the standard deviation of the population
- s/(n-1)
- б/√n

Question 8: If we have two normally distributed populations, then the sampling distribution of the difference of sample means will be:

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- Normal distribution ✓
- F-distribution
- t-distribution
- Chi-square distribution

Question 9: Which of the following must not be a consistent estimator?

- Sample mean
- Sample variance
- Sample median ✓
- Sample Proportion

Question 10: Given the following probability distribution:

P	**0**	**1/3**	**2/3**	**1**
f(p)	1/10	4/10	4/10	1/10

The mean of the sampling proportion will be:

- 15/30
- 1/15
- 1/20
- 10/20 ✓

Question 1: In general sample variance is a/an .....estimator?

- none of these
- Biased
- Consistent
- **Unbiased** ✔

Question 2: What is the rule of thumb for a sample size "n" to be considered large?

- **n=30** ✔

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- n<30
- n>30 or n=30
- None of the above

Question 3: Normal approximation to the binomial distribution is used when:

- None of the above
- np>5
- nq>5
- **Both of the above** ✔

Question 4: The standard error of the mean is the:

- **Standard deviation of the sampling distribution of sample means** ✔
- Sample standard deviation
- Standard mean of the sample
- Sample deviation of the sampling errors

Question 5: The method of Maximum Likelihood (ML) is used to find out:

- **Point estimates** ✔
- Random numbers
- Correlation coefficient
- Sample values

Question 6: The Maximum Likelihood Estimators (MLE) are …… and …… but not necessarily ……

- Unbiased, efficient, consistent
- Consistent, unbiased, efficient
- Unbiased, consistent, efficient
- **Consistent, efficient, unbiased** ✔

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Question 7: “A point estimate plus/minus a few times the standard error of that estimate”. This statement represents:

- Acceptance region
- Critical value
- **Confidence interval** ✔
- Critical region

Question 8: Method of Moments for estimation of parameters is based on the theory of:

- **Linear Estimation** ✔
- Non-linear estimation
- None of the above
- Both Linear and Non-linear

Question 9: An estimator is said to be efficient if it has:

- Unbiased estimate
- Both (a) & (b)
- None of these
- **Smallest Variance** ✔

Question 10: P 0 1 2 3
f(p) 1/20 9/20 9/20 1/20
The mean of the sampling proportion will be:

- 30/20
- **15/20** ✔
- 10/20
- 25/20

Question 1:
Poisson distribution is the limiting form of binomial distribution when n, the number of trials is large and p, the probability of success is:

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- very Small ✓
- Moderately large
- Moderately small
- very Large

Question 2:
The mean and the variance are equal in:

- The hypergeometric distribution.
- The Poisson distribution ✓
- The binomial distribution.
- All probability distributions.

Question 3:
A random experiment is one in which we repeat it a large number of times under similar conditions and it produces:

- Same results
- Different results ✓
- None of the above
- Independent results

Question 4:
E(X + Y) = ?

- None
- E(X + Y) = E(X) + E(Y). ✓
- E(X + Y) = E(X) E(Y).
- E(X + Y) = E(X) + E(Y).

Question 5:
The normal curve is:

- Symmetrical/Bell shaped ✓

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- Bimodal.
- Asymmetrical
- Multimodal.

Question 6:
Suppose 'X' is equal to the number of heads, when you throw a coin three times. The possible values of X are:

- 0,1,2,3 ✓
- 0,1,2,3,.....
- 1,2,3
- 1,2,3,.....

Question 7:
In sampling distribution of the sample proportion p is equal to:

- n/x
- x
- n
- x/n ✓

Question 8:
From the following table,
x | 0 | 1 | 2 | 3
f(x) | 0.26 | 0.49 | 0.22 | 0.02
E(2X+1) will be:

- 1.98
- 2.98 ✓
- 0.99
- 2

Question 9:
The number of students waiting at a bus stop in the morning and their probabilities are given below.
Where x = number of students waiting at the bus stop.

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x | 0 | 1 | 2 | 3 | 4
f(x) | 0.10 | 0.25 | 0.30 | 0.20 | 0.15
What is the probability that there are 2 (x=2) students waiting at the bus stop?

- 0.25
- 0.15
- 0.0
- 0.30 ✓

Question 10:
In a probability mass function, the sum of all the probabilities is equal to:

- 1 ✓
- 3
- 2
- 0

Question 1:

The number of students waiting at a bus stop in the morning and their probabilities are given below.
Where x = number of students waiting at the bus stop.

x	f(x)
0	0.10
1	0.25
2	0.30
3	0.20
4	0.15

Select the correct option:

☐ 0.01

☐ 0.10

☑ 0.25

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☐ 0.0

Question 2:

P[(X + Y) < 2] = ? , where x = 0, 1, 2 and y = 0, 1, 2.

Select the correct option:

☑ f(0,0) + f(0,1) + f(1,0)

☐ f(0,1) + f(0,2) + f(2,0)

☐ f(0,0) + f(1,1) + f(1,0)

☐ f(0,0) + f(1,1) + f(2,0)

Question 3:

The probability distribution of X is given below.

x	p(x)
0	0.73
1	?
2	0.06
3	0.04
4	0.01

What is the value of the missing probability?

Select the correct option:

☑ 0.16

☐ 0.06

☐ 0.54

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☐ 0.25

Question 4:

The probability mass function p(x) of a discrete random variable X is: P(0) = 0.10, P(1) = 0.15, P(2) = 0.45, P(3) = 0.05, and P(4) = 0.25.

What is the value of the cumulative distribution function for x = 2?

Select the correct option:

☑ 0.5

☐ 0.5

☐ 0.7

☐ 0.3

Question 5:

A correlation of -0.5 would indicate a scatterplot in which the slope is:

Select the correct option:

☐ curvilinear

☐ none of these

☐ downwards

☑ upwards

Question 6:

In a bivariate probability distribution of X and Y, if f(1,1) = 4/15, g(1) = 2/3, h(1) = 2/5, then:

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Select the correct option:

☐ None of these

☑ X and Y are statistically independent

☐ X and Y are not statistically independent

☐ Both of these

Question 7:

In hypergeometric distribution the following condition is violated:

Select the correct option:

☐ Both a) and b)

☑ Independence of trials

☐ None of the above

☐ Dependence of trials

Question 8:

In a bivariate distribution f(xy), when one random variable is discrete and the other is continuous, then we may call it:

Select the correct option:

☐ Co-discrete distribution

☐ Matching distribution

☐ Mixed distribution

☑ Paired distribution

Question 9:

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When a coin is tossed 3 times, the probability of getting 3 or less tails is:

Select the correct option:

☑ 1

☐ 1/2

☐ 3/5

☐ 0

Question 10:

What range of limit to be used for INTEGRATION of joint p.d.f to obtain its marginal pdf?

Select the correct option:

☐ -1 to 1

☐ 0 to +∞

☐ 0 to 1

☑ −∞ to +∞

Question 1:

In a discrete probability distribution, relation between mean and variance is:

Mean = Variance ✓

Mean > Variance

Difficult to tell

Mean < Variance

Question 2:

In a binomial experiment with n trials, the variable can take:

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2 values

4 values

n values ✓

3 values

Question 3:

Which type of data is used in the binomial distribution?

Both continuous and continuous

Discrete data only ✓

Qualitative data only

Continuous data only

Question 4:

The number of parameters in uniform distribution is/are:

1

2 ✓

0

4

Question 5:

Which of the following is true for the Poisson distribution?

Mean = Variance ✓

Mean < Variance

Mean > Standard deviation

Mean < Variance

Question 6:

According to empirical rule, how much data lies between $\mu - 3\sigma$ and $\mu + 3\sigma$?

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95.44% ✓

99.70%

80.28%

25%

Question 7:

In a hypergeometric distribution, the probability of success:

Just probability of failure

Remains constant from trial to trial

Does not remain constant from trial to trial ✓

Less than probability of failure

Question 8:

E(X + Y) = ?

E(X) - E(Y)

None

E(X) + E(Y) ✓

E(X) * E(Y)

Question 9:

Which of the following is the requirement of a uniform distribution?

Only two possible outcomes

A constant probability of success

Equally likely outcomes ✓

A fixed number of trials

Question 10:

The hypergeometric probability distribution has parameters:

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N, n

N, p

n, p

N, n, k ✓

Question 1:

Ten percent of the population is left-handed. In a class of 300 students, which of the following is the correct statement for "x" where x is at most left-handed students in the class?

Options:

P(x)

P(x ≥ 12)

P(x ≤ 12) ✓

P(x = 12)

Question 2:

If the first moment about origin is less than 0, then the distribution will be:

Options:

Positively Skewed

None of these

Negatively Skewed ✓

Symmetrical

Question 3:

Which property is true for the probability density function f(x)?

Options:

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f(x) is unbounded

f(x) = 1 for all x

f(x) = 0 for all x ✓

f(x) ≥ 0 for all x

Question 4:

If x = 3, then (2x + 3)2 - 1 = ...

Options:

2

3

15 ✓

6

Question 5:

Uniform distribution is a:

Options:

Mixed

Continuous Distribution ✓

Discrete distribution

None of the above

Question 6:

In a probability mass function, the sum of all the probabilities is equal to:

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Options:

0

2

1 ✓

3

Question 7:

Integrating the product of x and f(x) from range -∞ to +∞ gives us the:

Options:

First moment about origin ✓

Cumulative probability of x

First moment about mean

Total probability of x

Question 8:

From the following table:

x	0	1	2	3
P(X = x)	0.1	0.4	0.2	0.3

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Options:

0.97 ✓

0.12

0.38

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0.46

Question 9:

Which of the following is a characteristic of a binomial probability experiment?

Options:

P(Success) = P(Failure)

The result of one trial does not affect the probability of success on any other trial ✓

Each trial has more than two possible outcomes

Probability of success changes for each trial

Question 10:

If the second moment about origin is less than 3, the distribution will be:

Options:

Leptokurtic

None of these

Mesokurtic

Platykurtic ✓

1. From the above table, the expected value will be equal to:

a) 0.81

b) 0.79

c) 0.99 ✓

d) 0.71

2. The sum of the probabilities of bivariate probability function

f

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(
x
,
Y
)
f(x,Y) is 1. Then what will be the value of the sum of its marginal probability function
f
(
x
)
f(x)?

a) May and may not be one

b) Will be the average of two variables

c) Depends on the Y variable

d) Will be one ✓

3. Position of the normal curve changes along the horizontal axis due to:

a) Standard deviation ✓

b) Median

c) Mode

d) Mean

4. A uniform distribution has the following shape:

a) Rectangular ✓

b) Bi-modal

c) Bell-shaped

d) U-shaped

5. Which of the following statements is true about the normal distribution?

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a) It is a discrete probability distribution

b) It has a U-shaped probability density function

c) It is also known as the Poisson distribution

d) It is completely determined by its mean and variance ✓

6. Which of the following is true for the Poisson distribution?

a) mean > variance

b) mean = standard deviation

c) mean = variance ✓

d) mean > variance

7. What does it mean if the standard deviation is equal to zero?

a) All of the values in the data are same ✓

b) Mean = median = mode

c) All values of the data appear with the same frequency

d) The mean of the data is also zero

8. In a binomial experiment:

a) None of the above

b) Number of trials "n" is a variable

c) Number of trials "n" is not specified

d) Number of trials "n" is fixed ✓

9. Poisson distribution is the limiting form of binomial distribution when

n

n, the number of trials is large and

p

p, the probability of success is:

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a) very small ✓

b) Moderately large

c) very large

d) Moderately small

10. A discrete probability function

f

(

x

)

f(x) is always:

a) Negative

b) Non-negative ✓

c) Zero

d) One

1. If X and Y are independent random variables, then Cov (x, Y) =
    - 1
    - 0.5
    - ✓ 0
2. In hyper geometric distribution; N= number of units in the population, n = number of units in the sample and k = number of successes in the population, then Variance of hyper geometric distribution is defined[1] as:
    - nk/N ((N-1)/N)((N-n)/(N-1))
    - ✓ n.k/N ((N-k)/N)((N-n)/(N-1))
    - n.k/N ((N-k)/n)((N-n)/(N-1))
    - n.k/N ((N-k)/N)((N-n)/(N-k))
3. The location and shape of the normal curve is (are) determined by:
    - ✓ Mean & standard deviation
    - Mean & variance

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- Variance
- Mean

4. For a continuous random variable X, P(X = x) is:
   - 1
   - 0.5
   - undefined
   - ✓ 0
5. Match the binomial probability P(x< 23) with the correct statement.
   - P (there are more than 23 successes)
   - P(there are at most 23 successes)
   - ✓ P (there are fewer than 23 successes)
   - P (there are at least 23 successes)
6. For a binomial distribution, n= 10 & q= 0.6, the mean of the distribution is:
   - 0.6
   - 10
   - ✓ 6.0
7. Density function of normal distribution attains its maximum value when the value of random variable x is:
   - X = Standard Deviation
   - X = Variance
   - All above
   - ✓ X = Mean
8. In a binomial experiment:
   - Number of trials "n" is not specified
   - ✓ Number of trials "n" is a fixed
   - None of the above
   - Number of trials "n" is a variable
9. Given below is the probability distribution of random variable X. What is the value of "k"? x 0 1 2 3 4 p(x) .05 .02 .07 k .03

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- 1
- ✓ 0.20
- 0.02
- 0.002

10. In hyper geometric distribution formula, N= number of units in the population, n = number of units in the sample, and k is defined[2] as:
    - number of successes in sample
    - number of failures in population
    - number of failures in the sample
    - ✓ number of successes in population

The binomial distribution is negatively skewed when:

- p=q=1/2
- p=q
- pq
- pq
- ✔ **p=q=1/2**

Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. The distribution of Y is:

- Binomial with n = 5 and p = 0.5
- Poisson with u = .6
- ✔ **Binomial with n = 5 and p = 0.4**
- Binomial with n = 5 and p = 0.6

Ten percent of the population is left-handed. In a class of 100 students, which of the following is the correct statement for "there are at most 12 left-handed students in the class":

- ✔ **P(x ≤ 12)**
- P(x < 12)
- P(x > 12)
- P(x = 12)

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Which property is true for the probability density function f(x)?

- f(x) < 0 for all x
- f(x) > 0 for all x
- ✔ **f(x) ≥ 0 for all x**
- f(x) ≤ 0 for all x

Let X and Y are two random variables. Which property of expectation is true?

- E(x + Y) = E(X) + E(Y) - E(xY)
- E(X + Y) = 2 * {E(X) + E(Y)}
- E(X + Y) = 2E(X + Y)
- ✔ **E(X + Y) = E(X) + E(Y)**

A value of standard deviation is equal to 0, means:

- All the values in data are changing rapidly
- All the values are constant
- None of the above
- Both above
- ✔ **All the values are constant**

The range of the binomial distribution is:

- 0, 1, 2, ..., n
- ✔ **0, 1, 2, ..., n**
- n
- 0, 1, 2, ..., 10

What are the numbers of parameter of Hypergeometric distribution?

- ✔ **3**
- 2
- 1
- Undefined

x 0 1 2 3 f(x) 0.26 0.49 0.22 0.02 From the above table, the expected value will be equal to:

- 0.81

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- 0.79
- ✔ **0.99**
- 0.7

The number of telephone calls that pass through a switchboard has a Poisson distribution with mean equal to two per minute. The probability that no telephone call passes through the switchboard in a minute is:

- ✔ **0.1353**
- 0.2707
- 0.0517
- 0.0183

The hypergeometric random variable is a(an):

- Continuous variable
- Independent
- Undefined
- ✔ **Discrete variable**

The correlation coefficient:

- can be expressed as a percentage.
- all of these.
- ✔ **should be interpreted alongside the relevant scatterplot.**
- makes the scatterplot redundant.

The location parameter of normal distribution is:

- ✔ **Mean**
- Standard Deviation
- Variance
- Range

The uni-variate distribution obtained from a bi-variate distribution is known as:

- None of the above
- ✔ **Marginal Distribution**
- Both of the above

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- Joint Distribution

 If the second moment ratio is equal to 0, then the distribution will be:

- Negatively Skewed
- Leptokurtic
- Positively Skewed
- ✔ **Symmetrical**

 Poisson distribution is the limiting form of binomial distribution when n, the number of trials is large and p, the probability of success is:

- Moderately small
- ✔ **Very Small**
- Very Large
- Moderately large

 P(IXI < 1.5) =

- P(X = 15)
- P(X > 1.5)
- P(-15 < x < 15)
- ✔ **P(X < 1.5)**

 Asales firm receives different calls from clients. If someone says, today it will receive "at most 3 calls". How many calls will be received?

- ✔ **0, 1, 2, 3**
- 3, 4, 5, ...
- 1, 2, 3
- 0, 1, 2, 3, 5, ...

 If the second moment ratio is less than 3, the distribution will be:

- ✔ **Platykurtic**
- Leptokurtic
- None of these
- Mesokurtic

 If the second moment ratio is greater than 3, the distribution will be:

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- ✔ **Leptokurtic**
- None of these
- Mesokurtic
- Platykurtic

1. **Which of the following statements is an example of a continuous random variable?**
    - The number of credit hours you have this semester
    - All of the above
    - ✓ The depth of Lake Michigan at a randomly chosen point on the surface
    - The number of gas stations in Detroit
2. **A discrete probability function f(x) is always:**
    - ✓ Non-negative
    - Zero
    - Negative
3. **Match the binomial probability P(x> 0.5) with the correct statement.**
    - ✓ P(there are more than 0.50 successes)
    - P(there are fewer than 0.50 successes)
    - P(there are at least 0.50 successes)
    - P(there are at most 0.50 successes)
4. **In hyper geometric distribution the following condition is violated:**
    - Dependence of trials
    - None of the above
    - ✓ Independence of trials
    - Both a) and b)
5. **Let X and Y be two independent random variables then E(XY) =**
    - ✓ E(X) E(Y)
    - E(x)/E(Y)
6. **If the second moment ratio is greater than 3, the distribution will be:**

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- Platykurtic
- ✓ Leptokurtic
- Mesokurtic
- None of these

7. **The shape parameter of normal distribution is:**
   - ✓ Standard Deviation
   - Mode
   - Median
   - Mean
8. **If the total probability of joint probability distribution f(x,y) is ONE, then what will be the total probability of its marginal pdf g(x)?**
   - Half of the joint pdf
   - ✓ One
   - Less than one
   - Anywhere between zero and one
9. **According to the empirical rule, how much data lies between μ - 20 and μ + 20?**
   - 95.44%
   - 68.26%
   - 50%
   - ✓ 99.73%

1. For interpreting the value of t using a t-distribution table, the factors that must be known are:

- Whether hypothesis is one or two-tailed
- The significance level
- The degrees of freedom
- **All of the above**

2. If we are testing HO: P< P2(or) Pl - P2 < 0 against HI: Pl > P2(or) Pl - P2 > 0 and the tabulated value of Z = 1645 and Z calculated value = 2.63 then what will be the conclusion:

- Impossible to decide

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- **Reject Ho**
- None of the above
- Accept Ho

3. When we start hypothesis testing, we always assume that:

- HO is false
- Hlis true
- **HO is true**
- All are correct

4. To test the hypothesis about the difference of means for small samples, which test statistic will be used?

- **t test**
- Z-test
- F-test
- All of these

5. Alpha is the probability of:

- Type | error
- **All of these**
- Reject HO/HO is true
- Committing Type | error

6. Suppose HO: u1< u 2 HI :u1> u 2 and Critical value is = 1.645 If calculated value of Z = 1.88 then what will be your conclusion?

- **Reject Ho**
- Accept Ho
- None of the above
- Impossible to decide

7. If we are testing HO :P1 - P2 = 120 against HI: Pl - P2 > 120 (exceeds 120).Then rejection region will be:

- in Center
- on left side of Z=0
- **on right side of Z=0**

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- on both sides of Z=0

8. If we reject the true null hypothesis using t-distribution, we might be making:

- Type | error
- Acorrect decision
- Unpredictable
- Type II error

9. All other things remaining constant, a 99% confidence interval for a population mean p will

- be narrower than the 95% confidence interval for p.
- **be wider than the 95% confidence interval for p.**
- None of the above

10. For a particular hypothesis test, Alpha=0.09 and Beta=0.03, what is the value of Type | error?

- **0.09**
- 0.97
- 0.03
- 0.91

11. For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type II error?

- **0.15**
- 0.10
- 0.85
- 0.90

12. Mean of t-distribution will not exist when d.f(degree of freedom) will equal to:

- **2**
- 3

13. For the smaller values of v, the t-distribution will be

- than the standard normal distribution.
- **more spread out**
- same length
- less spread out

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14. When the population is normally distributed, population standard deviation is unknown and the sample size is n = 15, the confidence interval for the population mean is based on

- the Poisson Distribution.
- **the t distribution.**
- the Binomial distribution.
- the z (standard normal)distribution.

15. To test the hypothesis about the difference of means, which test we use for testing?

- **Both a and b**

16. Which is the most appropriate statistical technique used to test for significant differences between two group means from data gathered by a repeated measures design?

- **Paired t-test**
- Pearson correlation
- Student's t-test
- None of the above

17. The 'l-alpha' is the probability of:

- **Acceptance region**
- Type-l error
- Rejection region
- Type-ll error

18. In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to:

- **5**

19. The median of the t-distribution is

- **Zero**
- Does not exist
- Equals degrees of freedom

20. t-distribution is used to test the hypothesis about:

- **Mean**
- variance
- ratio of variance

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- Proportions

21. "The temperatures of patients were checked before and after given a certain medicine." This is an example of:

- **Pairing by design**
- Independent data
- Natural pairing of data
- Automatic grouping

22. Probability of making type Il error is reprented by:

- **Beta**
- a
- ba
- VB

23. Conventionally, the probability of making a type-I error is denoted by:

- **a (alpha)**
- 6 (sigma)
- & (beta)
- @ (theta)

24. How does the shape of the t-distribution change as the sample size increases?

- **It becomes more normal looking**
- It becomes broader
- It becomes flatter
- It becomes skewed

25. If we reject the true null hypothesis using t-distribution, we might be making:

- **Type I error**
- Acorrect decision
- Type Il error
- Unpredictable

26. For a sample pl = 0.2,nl= 100 and for second sample p2= 0.5 and n2= 200, the pooled estimator pc is:

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- **0.35**
- 0.30
- 0.40
- 0.45

27. For a sample pl= 0.4, nl = 100 and for second sample p2= 0.6 and n2= 100, the pooled estimator pc is:

- **0.50**
- 0.55
- 0.40
- 0.60

28. The proportion of working females in Pakistan is at most 0.40,the null hypothesis Ho<0.40 the alternative hypothesis HI is:

- **P>0.40**
- P=040
- P< 040

29. The critical region is computed from the value of:

- **1-alpha**
- Alpha
- 1-Beta
- Beta

Match the binomial probability P(x> 0.5) with the correct statement.

- **P(there are more than 0.50 successes)**
- **P(there are fewer than 0.50 successes)**
- **P(there are at most 0.50 successes)**
- **P(there are at least 0.50 successes)**

Correct Answer: P(there are at least 0.50 successes)

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2. The normal curve is:

- **Symmetrical/Bell shaped**
- **Bimodal.**
- **Multimodal.**
- **Asymmetrical**

Correct Answer: Symmetrical/Bell shaped

3. The number of trials in hypergeometric distribution is:

- **Large**
- **Not fixed**
- **Fixed**
- **Small**

Correct Answer: Fixed

4. Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

- **0.1923**
- **0.2132**
- **0.5133**
- **0.1713**

Correct Answer: 0.1713

5. In hyper geometric distribution; N = number of units in the population, n = number of units in the sample and k = number of successes in the population, then mean of hyper geometric distribution is defined as:

- **nk/N**
- **N/k**
- **nk/2N**
- **k/N**

Correct Answer: nk/N

6. In which of the following distributions is the probability of a success usually small?

- **Binomial**
- **Poisson**

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- **Hypergeometric**
- **All distribution**

Correct Answer: Poisson

7. The variance of the Hypergeometric distribution is:

- **{nk(N-k)(N-n)}/N*N(N-1)**
- **npq**
- **{nk(N-n)}/N(N-1)**

Correct Answer: {nk(N-k)(N-n)}/N*N(N-1)

8. Poisson distribution can be used to approximate the hyper geometric distribution when; n<0.05N, n>20 and:

- **p?0.05**
- **p<0.05**
- **p?0.10**
- **p<0.10**

Correct Answer: p < 0.05

9. Which type of data is used in the binomial distribution?

- **Both discrete and continuous**
- **Continues data only**
- **Discrete data only**
- **Qualitative data only**

Correct Answer: Discrete data only

10. The marginal p.d.f. of any continuous variables is obtained by:

- **Averaging**
- **Multiplying**
- **Integrating**
- **Dividing**

Correct Answer: Integrating

Question 1:

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Which of the following statements is true about the normal distribution?

- A. It is a discrete probability distribution
- B. **It is completely determined by its mean and variance**
- C. It is also known as the Poisson distribution
- D. It has a U-shaped probability density function

Question 2:

The range of hypergeometric distribution is:

- A. **0 to n or k (whichever is less)**
- B. 0 to n
- C. 0 to N
- D. 0 to k

Question 3:

The second moment-ratio of a particular continuous probability distribution is 2.144. What does this value tell us about the shape of the distribution?

- A. Flatter than normal distribution
- B. Normal distribution
- C. Neither very peaked nor flatter
- D. **Higher peak than the normal distribution**

Question 4:

In a normal distribution, how much area lies between μ - 3σ and μ + 3σ?

- A. 89%
- B. 70%
- C. 68%
- D. 99.73%

Question 5:

What should be the type of both variables in the joint probability distribution of X and Y?

- A. Both discrete
- B. Both continuous
- C. **Either both continuous or both discrete**

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- D. One continuous and the other discrete

Question 6:

Poisson distribution can be used to approximate the hypergeometric distribution when:

- A. $n < 0.05N$, $n > 20$
- B. **$p < 0.05$**
- C. $p < 0.10$
- D. $p > 0$

Question 7:

Which of the following is NOT applicable to a Poisson distribution?

- A. If $p = 0.02$ & $n = 300$
- B. If $p = 0.05$ & $n = 19$
- C. If $p = 0.01$ & $n = 200$
- D. **If $p = 0.03$ & $n = 500$**

Question 8:

According to the empirical rule, how much data lies between $\mu - \sigma$ and $\mu + \sigma$?

- A. 75.66%
- B. **95.44%**
- C. 99.73%
- D. 68.26%

Question 9:

For a good approximation of Poisson distribution to the binomial distribution, which of the following condition(s) is/are required?

- A. The probability, p, is close to 0.6 and the population size is large
- B. **The probability, p, is small and the sample size is large**
- C. The probability, p, is close to 0.5 and the sample size is large
- D. The population size is large relative to the sample size

Question 10:

When X and Y are _____ variables, then their joint probability function can be obtained by multiplying together their marginal probability functions.

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- A. Bivariate
- B. Normal
- C. Dependent
- D. **Independent**

Question 1:

- **Question:** We can obtain the individual probabilities of x and y from the joint probability function/ distribution of f(x,y). Such individual probabilities are known as
    - Options:
        - Separate probabilities
        - Independent probabilities
        - Bivariate probabilities
        - **Marginal probabilities** ✓

Question 2:

- **Question:** The number of trials in hypergeometric distribution is:
    - Options:
        - Large
        - **Fixed** ✓
        - Small
        - Not fixed

Question 3:

- **Question:** The hypergeometric probability distribution has parameters:
    - Options:
        - Nn
        - N,N-K,n
        - NP,n
        - **N,n,k** ✓

Question 4:

- **Question:** In a standard normal distribution area to the left of z = 0 is:
    - Options:

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        - 78%
        - 25%
        - **50%** ✓
        - 30%

Question 5:

- **Question:** The shape parameter of normal distribution is:
    - Options:
        - Mode
        - **Standard Deviation** ✓
        - Mean
        - Median

Question 6:

- **Question:** According to the empirical rule, how much data lies between μ - 2σ and μ + 2σ?
    - Options:
        - 99.73%
        - **95.44%** ✓
        - 50%
        - 68.26%

Question 7:

- **Question:** What are the numbers of parameters of the Hypergeometric distribution?
    - Options:
        - 3
        - **2** ✓

Question 8:

- **Question:** Which type of data is used in the binomial distribution?
    - Options:
        - Qualitative data only
        - Continuous data only

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        - **Discrete data only** ✓
        - Both discrete and continuous

Question 9:

- **Question:** Which of the following is the property of a hypergeometric experiment?
    - Options:
        - Successive trials are independent
        - **Sampling is performed without replacement** ✓
        - p remains constant from trial to trial
        - n is not fixed

Question 10:

- **Question:** The correlation coefficient:
    - Options:
        - makes the scatterplot redundant.
        - can be expressed as a percentage.
        - all of these.
        - **should be interpreted alongside the relevant scatterplot.** ✓

MCQs with Options and Correct Answers

Question 1:

- **Question:** The location and shape of the normal curve is (are) determined by:
    - **Options:**
        - Mean O
        - Variance
        - Mean & variance
        - Mean & standard deviation
        - O O O
    - **Correct Answer:** Mean & standard deviation ✓

Question 2:

- **Question:** Which of the following is the requirement of a uniform distribution?

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- **Options:**
  - Equally likely outcomes
  - A constant probability of success
  - A fixed number of trials
  - Only two possible outcomes
  - O O O
- **Correct Answer:** Equally likely outcomes ✓

Question 3:

- **Question:** E(x + Y) =?
  - **Options:**
    - O
    - None
    - E(x + Y) = E(x) E(Y).
    - E(X + Y) = E(x) + E(Y).
    - O O O
  - **Correct Answer:** E(X + Y) = E(x) + E(Y). ✓

Question 4:

- **Question:** Nature of the binomial random variable X is:
  - **Options:**
    - Continuous
    - Qualitative
    - Discrete
    - Quantitative
    - O O O
  - **Correct Answer:** Discrete ✓

Question 5:

- **Question:** Poisson distribution is the limiting form of binomial distribution when n, the number of trials is large and p, the probability of success is
  - **Options:**

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        - Moderately small
        - very Large
        - O
        - very Small
        - Moderately large
    - **Correct Answer:** very Small ✓

Question 6:

- **Question:** Two discrete r.v.'s X and Y are said to be statistically independent, if and only if, for all possible pairs of values (xi, Yj) the joint probability function f(x, y) can be expressed as the
    - **Options:**
        - O
        - Inverse
        - Product
        - Average
        - O O O
    - **Correct Answer:** Product ✓

Question 7:

- **Question:** Which of the following is a major property of hypergeometric distribution other than binomial distribution?
    - **Options:**
        - The experiment is repeated a fix number of time
        - It follows a continuous distribution
        - The successive trials are not independent
        - The outcome of each trial may be classified into one of two categories
        - O O O
    - **Correct Answer:** The successive trials are not independent ✓

Question 8:

- **Question:** The hypergeometric random variable is a(an):

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- **Options:**
  - Discrete variable
  - Independent
  - O
  - Continuous variable
  - Undefined
- **Correct Answer:** Discrete variable ✓

Question 9:

- **Question:** The mean of the continuous probability function f(x) can be obtained by
  - **Options:**
    - Integrating the x
    - O
    - Integrating the product xf(x)
    - Integrating the function f(x)
    - Multiplying the x by its function f(x)
    - O O O
  - **Correct Answer:** Integrating the product xf(x) ✓

Question 10:

- **Question:** P (IXI < 1.5) =
  - **Options:**
    - O
    - P(X = 1.5)
    - P(X > 1.5)
    - P(X < 1.5)
    - O O O
    - P(-15<X<15)
  - **Correct Answer:** P(-15<X<15) ✓

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1. In a random sample of 500 men from Lahore city, 300 are found to be smokers; the proportion of smokers is equal to:

Options:

- a. 0.4
- b. 0.5
- c. 0.6 (Correct)
- d. 0.3

2. As sample size goes up, what tends to happen to 95% confidence intervals?

Options:

- a. They become more narrow (Correct)
- b. They become more precise
- c. They become wider
- d. They become more precise and narrow

3. Suppose H0: p = 0, H1: p < 0, and Critical value is z = -2.33. If the calculated value of Z = -2.60, then what will be your conclusion?

Options:

- a. Accept Ho
- b. Reject Ho (Correct)
- c. Reject Ho
- d. Impossible to decide

4. The larger the n, the smaller the standard error, and so the narrower the _____.

Options:

- a. Level of significance
- b. None of the above
- c. Confidence level (Correct)

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5. t-distribution can be used for testing about:

Options:

- a. Population mean (Correct)
- b. Population variance
- c. Difference between population proportions
- d. Population proportion

6. If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

Options:

- a. None of these
- b. P > 3
- c. P < 3
- d. P = 3

7. If we reject the true null hypothesis using t-distribution, we might be making:

Options:

- a. Unpredictable Type II error
- b. Type I error (Correct)
- c. A correct decision

8. For a particular hypothesis test, Alpha = 0.04 and Beta = 0.06, what is the value of Type II error?

Options:

- a. 0.04
- b. 0.06
- c. 0.94 (Correct)
- d. 0.96

9. The term 1 - β is called:

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Options:

- a. Power of the test (Correct)
- b. Level of the test
- c. Size of the test
- d. Critical region

10. The probability of Type II error is equal to:

Options:

- a. Alpha
- b. 1 - Alpha
- c. Beta (Correct)
- d. 1 – Beta

1. Q: ß is the probability of:

- A: Accept H0/H0 is true (Correct Option)

2. Q: For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type I error?

- A: 0.10 (Correct Option)

3. Q: If we are testing H0 : u1 - u2 = 10 against H1: u1 - u2 < 10. Then the rejection region will be:

- A: On the left side of Z=0 (Correct Option)

4. Q: The Test-statistic is used to decide whether to _____ a null-hypothesis.

- A: Accept or Reject (Correct Option)

5. Q: We can apply t-distribution to test the hypothesis when:

- A: Population variance is unknown and Sample is small (Correct Option)

6. Q: T-distribution is applicable in case of:

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- A: Independent samples (Correct Option)

7. Q: _____ are equivalent.

- A: Type-I error and Level of significance (Correct Option)

8. Q: For a particular hypothesis test, Alpha=0.09 and Beta=0.03, what is the value of Type I error?

- A: 0.09 (Correct Option)

9. Q: If a significance level of 10% is used rather than 5%, the null hypothesis is:

- A: Less likely to be rejected (Correct Option)

10. Q: Suppose H0 : u1 = u2 and H1: u1 > u2, and the critical value is z=1.645. If the calculated value of Z=1.32, then what will be your conclusion?

- A: Accept H0 (Correct Option)

1. For the test of hypothesis Ho: μ1 = μ2 and H1: μ1 < μ2, the critical region at 0.05 level of significance and n > 30 is:

- A) Z < -1.645
- B) Z > 1.645
- C) Z less and equal 1.96
- D) Z > 1.96
- **Correct Answer: A) Z < -1.645**

2. The t-distribution can never become narrower than:

- A) F-distribution
- B) Chi-square distribution
- C) Exponential distribution
- D) Standard normal distribution
- **Correct Answer: D) Standard normal distribution**

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3. An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is a 95% confidence interval for the mean assembly time?

- A) (17.1123, 24.7277)
- B) (17.1123, 22.7277)
- C) (14.1123, 22.7277)
- D) (17.1123, 30.7277)
- **Correct Answer: B) (17.1123, 22.7277)**

4. Pairing is done only in case of …….observations.

- A) Dependent
- B) Small
- C) Large
- D) Independent
- **Correct Answer: A) Dependent**

5. If a significance level of 10% is used rather than 5%, the null hypothesis is:

- A) Just as likely to be rejected
- B) More likely to be rejected
- C) None of these
- D) Less likely to be rejected
- **Correct Answer: D) Less likely to be rejected**

6. If we are testing H0: P1 < P2 (or) P1 - P2 < 0 against H1: P1 > P2 (or) P1 - P2 > 0 and the tabulated value of Z = 1.645 and Z calculated value = 2.63, then what will be the conclusion:

- A) Impossible to decide
- B) Accept Ho
- C) Reject Ho
- D) None of the above

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- **Correct Answer: C) Reject Ho**

7. Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

- A) θ (theta)
- B) σ (sigma)
- C) α (alpha)
- D) ß (beta)
- **Correct Answer: C) α (alpha)**

8. The density of the t-distribution reaches its maximum at t=_____:

- A) 0
- B) 12
- C) 3
- D) 4
- **Correct Answer: A) 0**

9. If a significance level of 1% is used rather than 5%, the null hypothesis is:

- A) Less likely to be rejected
- B) Just as likely to be rejected
- C) More likely to be rejected
- D) None of these
- **Correct Answer: A) Less likely to be rejected**

10. The critical region is computed from the value of:

- A) 1 - Beta
- B) Alpha
- C) Beta
- D) 1 - alpha

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- **Correct Answer: B) Alpha**

MCQ 1:

For a particular hypothesis test, α=0.05 and β=0.05. The power of the test is equal to:

Options:

- A. 0.90
- B. 0.95 (Correct Option)
- C. 0.05
- D. 0.50

MCQ 2:

How can we interpret the 90% confidence interval for the mean of the normal population?

Options:

- A. There are 90% chances of falling the true value of the parameter. (Correct Option)
- B. There are 100% chances of falling the true value of the parameter.
- C. There are 10% chances of falling the true value of the parameter.
- D. All are correct.

MCQ 3:

Conventionally, the probability of making a type-II error is denoted by:

Options:

- A. Θ (theta)
- B. α (alpha)
- C. б (sigma)
- D. ß (beta) (Correct Option)

MCQ 4:

The proportion of working females in Pakistan is at most 0.40. The null hypothesis Ho: P ≤ 0.40. The alternative hypothesis H1 is:

Options:

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- A. P ≠ 0.40 (Correct Option)

- B. P = 0.40

- C. P < 0.40

- D. P > 0.40

MCQ 5:

The degrees of freedom associated with paired t-test is:

Options:

- A. n - 1 (Correct Option)

- B. n1 + n2 - 4

- C. n1 - 2 and n2 - 2

- D. n - 2

MCQ 6:

Accept H0 when H0 is false is the _____.

Options:

- A. None of the above

- B. Correct decision

- C. Type I error

- D. Type II error (Correct Option)

MCQ 7:

For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type II error?

Options:

- A. 0.85

- B. 0.90 (Correct Option)

- C. 0.10

- D. 0.15

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MCQ 8:

Suppose H0: u1 = u2 and H1: u1 < u2. The critical value is z= -1.645. If the calculated value of Z= -2.08, then what will be your conclusion?

Options:

- A. Reject Ho (Correct Option)

- B. None of the above

- C. Impossible to decide

- D. Accept Ho

MCQ 9:

Interval estimation and confidence interval are:

Options:

- A. Independent

- B. Opposite

- C. Same (Correct Option)

- D. Different

MCQ 10:

In hypothesis testing, suppose the critical value is |Z| > 2.33. This value shows that the applied test is _____?

Options:

- A. Two-tailed test (Correct Option)

- B. Left-tailed test

- C. One-tailed test

- D. Right-tailed test

1. "A failed student was passed by the examiner," is an example of:

   - Type-II error

   - Type-III error

   - Type-IV error

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- Type-I error

- Correct Option: Type-II error

2. Critical region is also known as:

- Rejection region

- None of these

- Neither a and b

- Acceptance Region

- Correct Option: Rejection region

3. Mean of t-distribution will not exist when d.f (degree of freedom) will equal to:

- 1

- 2

- 3

- 4

- Correct Option: 1

4. The proportion of females in Pakistan is at most 0.52, the alternative hypothesis H1 is:

- P is less than & equal to 0.52

- P is equal to 0.52

- P is greater than 0.52

- P is greater than and equal to 0.52

- Correct Option: P is less than & equal to 0.52

5. For the smaller values of v, the t-distribution will be _____ than the standard normal distribution.

- None of these

- Less spread out

- Same length

- More spread out

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- Correct Option: More spread out

6. The term 1- β is called:

- Size of the test
- Critical region
- Level of the test
- Power of the test
- Correct Option: Power of the test

7. The proportion of males in Pakistan is at least 0.48, the Null Hypothesis is $P \geq 0.48$, then the alternative hypothesis H1 is:

- $P < 0.48$
- $P \leq 0.48$
- $P \geq 0.48$
- $P = 0.48$
- Correct Option: $P < 0.48$

8. To test the hypothesis about the difference of means for small samples, which test statistic will be used?

- F-test
- t-test
- Z-test
- All of these
- Correct Option: t-test

9. If we are testing H0: P1 - P2 = 120 against H1: P1 - P2 > 120 (exceeds 120). Then the rejection region will be:

- In Center
- On both sides of Z = 0
- On the left side of Z = 0

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- On the right side of Z = 0

- Correct Option: On the right side of Z = 0

10. t-distribution can be used for testing about:

- Population variance

- Population proportion

- Difference between population proportions

- Population mean

- Correct Option: Population mean

1. Confidence interval is also known as:

- A. Level of Significance

- B. Point estimator

- C. None of these

- **D. Confidence interval**

2. To test the hypothesis about the difference of means for large samples, what test statistic will be used?

- A. Chi-square

- B. 1

- C. 2

- **D. Z (Z-score)**

3. A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

- A. Hypothesis

- B. Hypothesis

- C. Hypothesis

- **D. Confidence interval**

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4. Suppose H0: p = 0 H1: p < 0 and Critical value is z = - 2.33. If calculate the value of Z = - 2.60, then what will be your conclusion?

- **A. Reject Ho**
- B. Accept Ho
- C. None of the above
- D. Impossible to decide

5. Pairing is done only in case of …….observations.

- A. Dependent
- B. Small
- C. Large
- **D. Independent**

6. If we are testing H0 :P1 - P2 = 120 against H1 : P1 - P2 > 120 (exceeds 120), then the rejection region will be:

- A. on both sides of Z=0
- B. on the left side of Z=0
- C. in the center
- **D. on the right side of Z=0**

7. For a sample p1= 0.4, n1 = 100 and for the second sample p2= 0.6 and n2= 100, the pooled estimator pc is:

- A. 0.500
- **B. 0.400**
- C. 0.600
- D. 0.55

8. The level of significance is also called:

- A. Type II error
- B. Power of the test

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- **C. Type I error**

- D. Acceptance region

9. In a t-distribution, for smaller values of v (degrees of freedom), the t-distribution will be _____ than the standard normal distribution.

- A. less spread out

- **B. more spread out**

- C. less spread out

- D. None of these

10. In a t-distribution:

- A. Mean > Median < Mode

- B. Median < Mode < Mean

- **C. Mean = median = mode**

- D. None of these

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Question 1: If

b

(

x

,

7

,

0.30

)

b(x,7,0.30), the variance of this distribution is:

( ) 1.74

(✓) 1.44

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( ) 1.77

( ) 1.47

( ) 0.30

Question 2: The normal curve is:

( ) Bimodal

(✓) Symmetrical/Bell shaped

( ) Asymmetrical

( ) Multimodal

Question 3: The number of telephone calls that pass through a switchboard has a Poisson distribution with mean equal to two per minute. The probability that no telephone call passes through the switchboard in a minute is:

( ) 0.0183

( ) 0.0517

( ) 0.2707

(✓) 0.1353

Question 4: In a bivariate probability distribution of X and Y, if

E

(

X

)

=

3

E(X)=3 and

E

(

Y

)

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=

5

E(Y)=5, then

E

(

X

−

Y

)

E(X−Y) is:

( ) 2

(✓) -2

( ) None of these

( ) 15

Question 5: Which of the following is NOT applicable to a Poisson distribution?

( ) IF P = 0.03 & n = 500

( ) IF P = 0.02 & n = 300

(✓) IF P = 0.5 & n = 19

( ) IF P = 0.01 & n = 200

Question 6: Match the binomial probability

P

(

x

≥

23

)

P(x≥23) with the correct statement:

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( ) P (there are more than 23 successes)

( ) P (there are fewer than 23 successes)

(✓) P (there are at least 23 successes)

( ) P (there are at most 23 successes)

Question 7: If we have

$f(x) = 4x, 0 \le x \le 1$

f(x)=4x,0≤x≤1, then

$f(x)$

f(x) is:

( ) Probability distribution

( ) Distribution function

( ) Continuous random variable

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(✓) Probability density function

Question 8: If the second moment ratio is less than 3, the distribution will be:

(✓) Leptokurtic

( ) Platykurtic

( ) Mesokurtic

( ) None of these

Question 9: In a binomial experiment, successive trials are:

( ) None

( ) Dependent

(✓) Independent

( ) Both

Question 10: If

E

(

X

)

=

2

E(X)=2 and

E

(

Y

)

=

5

E(Y)=5, then

E

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(

X

Y

)

E(XY) is equal to:

( ) 7

( ) 5

( ) 3

(✓) 10

1. If we reject the true null hypothesis using the t-distribution, we might be making:

Options:

- A. Unpredictable

- B. A correct decision

- C. Type I error

- D. Type II error

Correct Option: C. Type I error

2. For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type I error?

Options:

- A. 0.85

- B. 0.10

- C. 0.90

- D. 0.15

Correct Option: B. 0.10

3. For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type II error?

Options:

- A. 0.10

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- B. 0.85

- C. 0.90

- D. 0.15

Correct Option: D. 0.15

4. An industrial designer wants to determine the average amount of time it takes an adult to assemble an “easy to assemble” toy. What is a 95% confidence interval for the mean assembly time?

Options:

- A. (14.1123, 22.7277)

- B. (17.1123, 22.7277)

- C. (17.1123, 24.7277)

- D. (17.1123, 30.7277)

Correct Option: B. (17.1123, 22.7277)

5. For a particular hypothesis test, Alpha=0.05 and Beta=0.15. What is the level of significance?

Options:

- A. 0.05

- B. 0.85

- C. 0.15

- D. 0.95

Correct Option: A. 0.05

6. The proportion of working females in Pakistan is at most 0.40. The alternative hypothesis H1 is:

Options:

- A. $P \neq 0.40$

- B. $P = 0.40$

- C. $P < 0.40$

- D. $P > 0.40$

Correct Option: A. $P \neq 0.40$

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7. The term 1 - β is called:

Options:

- A. Power of the test
- B. Critical region
- C. Size of the test
- D. Level of the test

Correct Option: A. Power of the test

8. The degrees of freedom associated with a paired t-test are:

Options:

- A. n - 2
- B. n1 + n2 - 4
- C. n1 - 2 and n2 - 2
- D. n - 1

Correct Option: D. n - 1

9. If a significance level of 5% is used rather than 1%, the null hypothesis is:

Options:

- A. More likely to be rejected
- B. None of these
- C. Just as likely to be rejected
- D. Less likely to be rejected

Correct Option: C. Just as likely to be rejected

10. If we have the following information: P = 0.16, q = 0.84, Z = 1.96, and e = 0.05, then the sample size will be:

Options:

- A. 207

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- B. 200

- C. 199

- D. 205

Correct Option: A. 207

1. t-distribution can be used to find the confidence interval for:

Options:

A) Population mean

B) Population proportion

C) Difference between population proportions

D) Population variance

Correct Option: A) Population mean

2. To test the hypothesis about the difference of means for large samples, what test statistic will be used?

Options:

A) F-test

B) Z-test

C) Chi-square

D) t-test

Correct Option: B) Z-test

3. Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income; the null hypothesis will be:

Options:

A) MeanA > MeanB

B) MeanA < MeanB

C) MeanA ≥ MeanB

D) MeanA ≤ MeanB

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Correct Option: D) MeanA ≤ MeanB

4. The term 1- β is called:

Options:

A) Critical region

B) Size of the test

C) Power of the test

D) Level of the test

Correct Option: C) Power of the test

5. For interpreting the value of t using a t-distribution table, the factors that must be known are:

Options:

A) The degrees of freedom

B) Whether the hypothesis is one or two-tailed

C) All of the above

D) The significance level

Correct Option: C) All of the above

6. To test the hypothesis about the difference of means for small samples, which test statistic will be used?

Options:

A) All of these

B) Z-test

C) F-test

D) t-test

Correct Option: D) t-test

7. Which is the most appropriate statistical technique used to test for significant differences between two group means from data gathered by a repeated measures design?

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Options:

A) Pearson correlation

B) Paired t-test

C) Student's t-test

D) None of the above

Correct Option: B) Paired t-test

8. If we are testing H0 : P1 < P2 (or) P1 - P2 < 0 against H1 : P1 > P2 (or) P1 - P2 > 0 and the tabulated value of Z = 1.645 and Z calculated value = 2.63, then what will be the conclusion?

Options:

A) Reject H0

B) None of the above

C) None of the above

D) Accept H0

Correct Option: A) Reject H0

9. In the Binomial distribution, the sample size is considered to be sufficiently large if both np and nq are greater than or equal to:

Options:

A) 3

B) 15

C) 5

D) 10

Correct Option: D) 10

10. The t-distribution is applicable for testing about the population mean when the population standard deviation is:

Options:

A) Greater than 40

B) Known

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C) Less than 40

D) Unknown

Correct Option: D) Unknown

1. If we are testing H0: mean = 50 against H1: mean < 50, then the rejection region will be on:

- a) on both sides
- b) on the left side of the mean
- c) in the center
- d) on the right side of the mean
- Correct option: b) on the left side of the mean

2. The level of significance is also called:

- a) Acceptance region
- b) Type I error
- c) Type II error
- d) Power of test
- Correct option: b) Type I error

3. Suppose H0: p = 0 and H1: p < 0, and the critical value is z = -2.33. If the calculated value of Z = -2.60, then what will be your conclusion?

- a) Accept H0
- b) Impossible to decide
- c) Reject H0
- d) None of the above
- Correct option: c) Reject H0

4. The test statistic to test the U1 = U2 (U represents the mean of population) for a normal population when n > 30 is:

- a) Z-Test

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- b) F-Test

- c) T-Test

- d) All of Above

- Correct option: a) Z-Test

5. For a particular hypothesis test, Alpha = 0.10 and Beta = 0.15. What is the value of Type I error?

- a) 0.15

- b) 0.90

- c) 0.85

- d) 0.10

- Correct option: d) 0.10

6. For a particular hypothesis test, Alpha = 0.04 and Beta = 0.06, what is the value of Type II error?

- a) 0.06

- b) 0.94

- c) 0.96

- d) 0.04

- Correct option: a) 0.06

7. Suppose H0: u1 = u2 and H1: u1 < u2, and the critical value is z = -1.645. If the calculated value of Z = -2.08, then what will be your conclusion?

- a) Impossible to decide

- b) None of the above

- c) Reject H0

- d) Accept H0

- Correct option: c) Reject H0

8. If we draw a sample of size ‘n’ from a population, then the sample is called a "large sample" if:

- a) n > 40

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- b) n > 25
- c) n > 30
- d) n > 50
- Correct option: c) n > 30

9. A confidence interval has a specified probability _____ of containing the true value of the parameter.

- a) 1 - α
- b) 1 + α
- c) α
- d) α/2
- Correct option: a) 1 - α

10. An alternative hypothesis is generally denoted by:

- a) H1
- b) H3
- c) H0
- d) H2
- Correct option: a) H1

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